



4948 - Flux-calibrating NIRSpec MSA data, finally

Cycle: 3, Proposal Category: AR

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Prof. Michael Maseda (PI)	University of Wisconsin - Madison
Dr. Anna G de Graaff (CoI) (ESA Member)	Max Planck Institute for Astronomy
Dr. Joel Leja (CoI)	The Pennsylvania State University
Dr. Themiya Nanayakkara (CoI)	Swinburne University of Technology
Prof. Karl Glazebrook (CoI)	Swinburne University of Technology
Dr. Benjamin D. Johnson (CoI)	Harvard University
Dr. Gwen C. Rudie (CoI)	Carnegie Institution of Washington
Prof. Allison L. Strom (CoI)	Northwestern University
Dr. Jarle Brinchmann (CoI) (ESA Member)	Centro de Astrofisica da Universidade do Porto
Pablo Arrabal Haro (CoI)	NOIRLab - (AZ)

OBSERVATIONS

ABSTRACT

NIRSpec has promised to revolutionize our ability to understand the growth and evolution of galaxies across cosmic time with sensitive spectroscopic observations far surpassing the capabilities of other instruments on other facilities. The design of its microshutter array (MSA), while allowing for simultaneous spectroscopy of more than 150 galaxies, means that flux calibration for resolved sources is extremely challenging due to the regular aperture size but spatially and spectrally varying point spread function. For a typical $z \sim 2$ galaxy, the STScI pipeline can incorrectly calibrate the absolute flux of the spectrum by a factor of 5 or more, depending on the wavelength and the centering of the source inside the aperture. In order to permit science that requires accurate flux information (star formation rates, metallicities, dust attenuation, stellar population information; i.e. most of the extragalactic science one would want to perform with NIRSpec), we will develop open-source software designed to calibrate the spectra taken with the MSA. We outline two methods which use morphological and/or photometric information for a galaxy combined with detailed

JWST Proposal 4948 (Created: Thursday, February 29, 2024 at 6:01:11 PM Eastern Standard Time) - Overview

models of the NIRSpec instrument in order to accurately determine the intrinsic flux level of a spectrum. With the newest calibration data taken in Cycles 1 and 2 as well as non-MSA spectroscopy, we will validate our methods across the full population of high-z galaxies. We will calibrate all Treasury and ERO MSA data and release the spectra to the community. Calibrating the flux level for NIRSpec MSA spectroscopy is crucial for the success of the mission and cannot be done correctly for most galaxies by the STScI pipeline.

OBSERVING DESCRIPTION